

YASAWA-I-RARA, Fiji

WMO: 916600

Lat: 16.700S

Lon: 177.583E

Elev: 164

StdP: 14.61

Time Zone: 12.00 (PCF)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	70.7	71.8	53.5	61.3	77.0	59.4	76.0	75.5	31.2	74.9	29.4	75.9	14.3	130	0.500

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.1	91.8	81.4	90.3	81.0	88.9	80.6	86.6	87.6	84.8	87.1	83.6	86.3	8.4	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
86.4	194.4	87.0	84.3	181.1	85.7	82.8	172.0	84.9	51.7	87.3	49.3	87.0	47.9	86.9	92.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.8	25.5	23.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.6	83.0	83.6	82.8	81.7	80.5	79.2	77.8	77.5	78.6	79.3	80.8	82.2
	DBStd	2.89	2.33	2.39	1.87	1.82	1.97	2.17	2.02	2.39	2.06	2.00	1.94	1.91
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11158	1023	941	1018	951	945	875	861	854	859	909	924	998
	CDD65	5683	558	521	553	501	480	425	396	389	409	444	474	533
	CDH74	51538	6285	6004	6107	5111	4186	3160	2311	2262	2721	3400	4372	5619
(14) Wind	WSAvg	12.3	11.4	10.2	9.9	12.3	11.7	12.4	13.5	14.5	12.8	13.1	12.0	13.0
	PrecAvg	81.2	12.1	10.6	11.0	8.4	5.7	4.1	2.6	2.9	3.7	5.4	6.2	8.5
	PrecMax	107.6	33.1	18.9	19.3	18.3	13.4	11.4	6.1	8.0	9.1	11.1	15.8	16.1
	PrecMin	51.3	2.9	1.4	3.3	1.2	0.6	0.7	0.2	0.3	0.7	0.6	2.0	1.6
	PrecStd	14.9	6.3	4.0	4.0	4.5	3.1	2.8	1.4	1.9	2.0	2.8	3.4	3.2
	DB	95.3	95.9	91.7	90.3	89.8	88.1	87.2	86.3	87.9	88.6	90.2	91.3	
	MCWB	82.6	82.2	82.2	80.0	80.5	78.3	76.2	77.7	78.1	78.3	78.5	79.2	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	91.6	92.3	89.9	88.4	87.7	86.0	84.6	84.4	85.6	86.3	87.9	89.1	
	2% MCWB	82.6	80.9	82.5	80.7	79.4	77.4	76.2	76.7	77.8	78.2	77.8	79.2	
	5% DB	89.5	90.3	88.6	86.8	85.7	84.4	83.0	83.0	84.1	84.7	86.3	87.8	
	5% MCWB	81.1	81.1	82.1	80.5	78.6	76.9	76.1	76.0	76.7	77.3	78.2	79.4	
	10% DB	87.6	88.3	87.2	85.5	84.1	82.8	81.3	81.4	82.7	83.4	84.9	86.4	
	10% MCWB	80.8	81.0	81.6	79.9	78.1	76.5	75.6	74.9	75.8	76.4	78.1	79.3	
	WB	89.5	88.3	88.4	86.3	86.7	82.6	81.9	81.9	83.5	82.9	84.2	86.6	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% MCDB	90.2	89.0	88.6	86.7	87.1	83.9	82.8	83.2	84.0	84.2	85.8	87.1	
	2% WB	86.0	84.8	85.6	83.7	83.2	80.8	80.0	79.6	80.1	80.7	82.3	84.0	
	2% MCDB	87.3	89.0	87.3	85.3	84.1	82.3	81.1	81.8	83.0	83.9	84.3	85.7	
	5% WB	83.7	83.3	83.9	82.3	81.0	79.3	78.8	78.1	78.5	79.2	80.9	82.5	
	5% MCDB	87.3	87.6	87.0	84.2	82.5	81.3	80.2	80.7	81.9	82.2	83.2	85.3	
	10% WB	82.3	82.2	82.6	81.1	79.6	78.2	77.5	76.8	77.3	78.2	79.8	81.3	
	10% MCDB	85.8	85.9	85.6	83.4	81.9	80.5	79.1	79.2	80.2	81.0	82.1	84.0	
(35) Mean Daily Temperature Range	MDBR	8.9	9.1	8.9	7.7	7.7	7.2	7.4	7.4	8.2	8.0	8.1	8.3	
	5% DB MCDBR	12.4	12.3	10.8	9.8	10.4	9.6	9.8	9.5	10.1	10.1	10.2	10.2	
	5% DB MCWBR	6.3	6.2	6.4	5.9	5.8	4.6	4.9	5.4	5.0	4.8	4.7	5.3	
	5% WB MCDBR	10.9	10.4	10.0	8.0	8.4	7.1	7.1	8.3	8.8	8.4	8.0	8.8	
	5% WB MCWBR	7.2	6.4	7.3	6.7	6.9	4.8	5.1	6.4	5.7	4.9	5.5	6.1	
(40) Clear-Sky Solar Irradiance	taub	0.387	0.384	0.375	0.375	0.372	0.371	0.366	0.366	0.378	0.382	0.382	0.382	
	taud	2.632	2.648	2.689	2.677	2.653	2.637	2.640	2.643	2.602	2.611	2.632	2.645	
	Ebn at Noon	304	302	298	287	276	271	276	285	292	298	303	305	
	Edh at Noon	32	31	29	28	27	26	27	28	31	32	32	32	
(44) All-Sky Solar Radiation	RadAvg	1995	1887	1762	1567	1395	1308	1383	1553	1771	1922	1985	2016	
	RadStd	212	172	146	119	117	81	69	136	115	131	195	101	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	+0.32	N/A	N/A	N/A	+0.36	+0.43	N/A	N/A	+121	+119

Nomenclature: See separate page